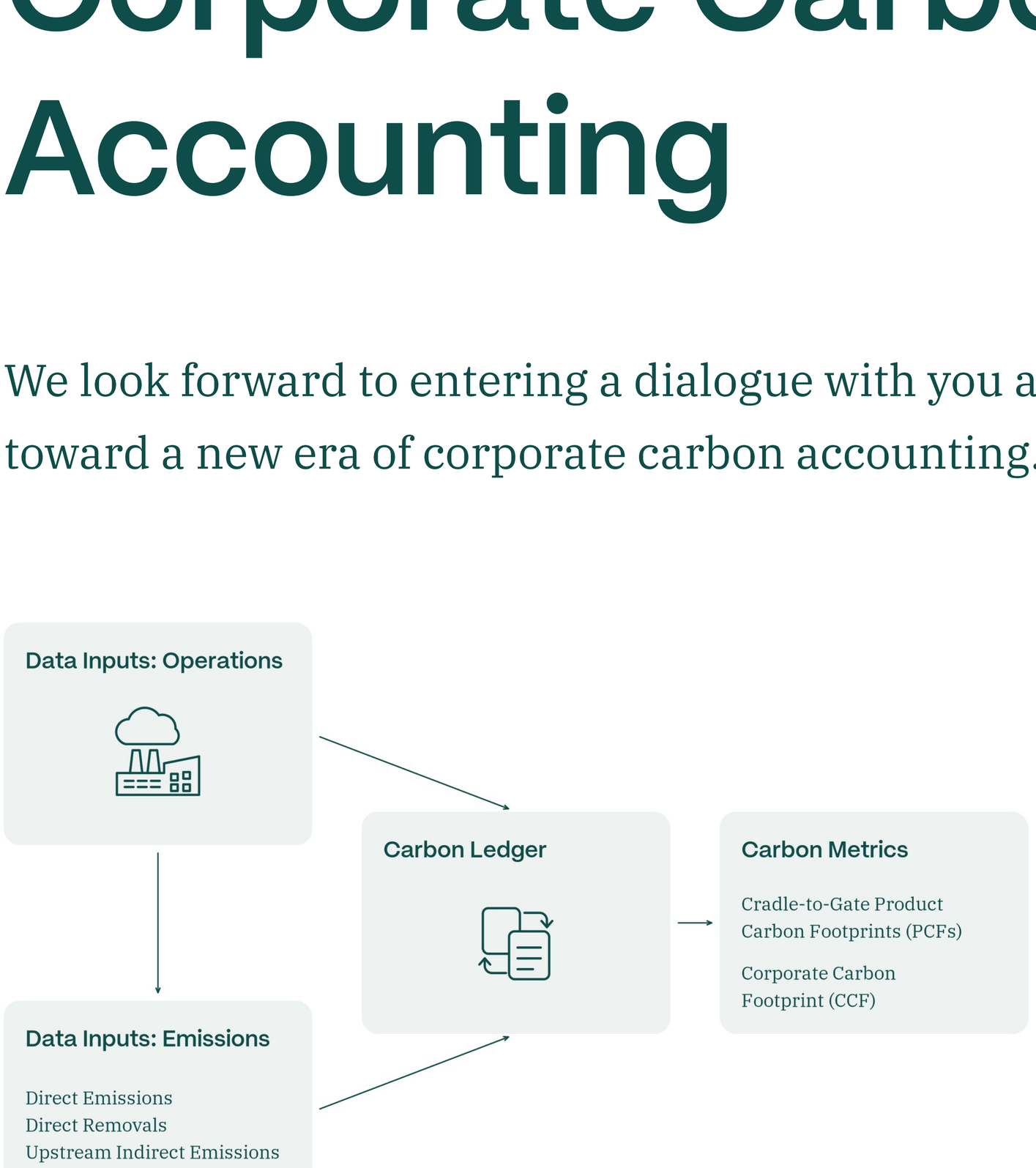
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Foundations of Integrated Corporate Carbon Accounting

We look forward to entering a dialogue with you as we work toward a new era of corporate carbon accounting.



One set of data inputs, one ledger, both metrics.

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Published

October 2025, revised December 2025

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C. Ernst · S. Reichelstein · G. Ormazabal · T. Sellhorn · The Carbon Accounting Standards Initiative (CASI) · October 2025, revised December 2025

Executive Summary

Businesses in a wide range of industries are increasingly considered responsible not only for their own direct carbon emissions but also for the emissions embodied in their sales products. This whitepaper describes the structure of Integrated Corporate Carbon Accounting (ICCA) systems that track an entity's direct (Scope 1) and indirect (Scope 2 and upstream Scope 3) emissions over time in order to calculate key carbon emission metrics.

Completeness

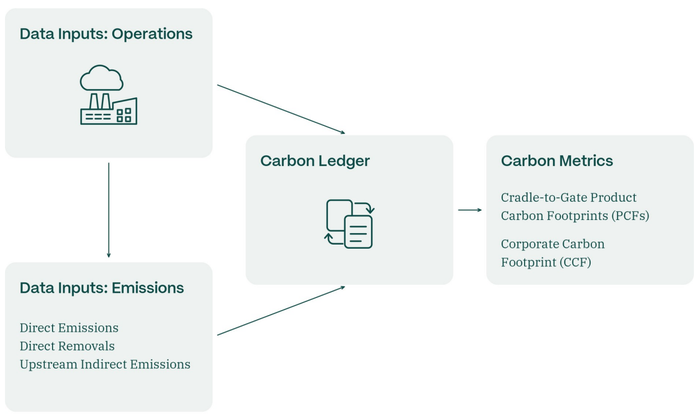
Companies account and assume responsibility for their own direct (Scope 1) emissions and the indirect emissions embodied in acquired inputs and services (Scope 2 and upstream Scope 3) during an accounting period. An ICCA system is therefore complete if the carbon ledger records all direct and upstream indirect emissions corresponding to incoming goods and services received by the business.

Accuracy

ICCA systems become more accurate as data inputs about indirect supply chain emissions are increasingly based on primary data that represent the actual emissions incurred in the supply chain. Accuracy will therefore improve as more companies in an entity's supply network adopt their own ICCA systems. Recent contributions have proposed to quantify the level of accuracy by means of a Data Quality Scoring (DQS) index.

Timely Recognition

Direct (Scope 1) emissions and direct removals are recorded in the carbon ledger for the accounting period in which they occur. Indirect emissions are recorded in the carbon ledger in the same time period in which customers acquire (take control of) goods and services. The carbon ledger must include accounts for carbon stock variables that capitalize any incoming emissions that are not recorded directly in the current CCF metric.



One set of data inputs, one ledger, both metrics.

Figure 3 Schematic for Integrated Corporate Carbon Accounting: data inputs, the carbon ledger and the resulting carbon metrics.